

BASEL TAREK OSMAN

Data Scientist | Machine Learning Engineer | Data Analyst

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PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer and Data Scientist with strong foundation in building end-to-end predictive models, data pipelines, and interactive analytical dashboards. Kaggle Expert with proven expertise in Python, Scikit-learn, and Natural Language Processing. Experienced in delivering data-driven solutions through internships at National Telecommunication Institute (NTI), Orange Digital Center, and Sprints. Specialized in transforming complex datasets into actionable business insights and developing practical AI solutions for real-world business problems.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Dart, TypeScript

Data Science Libraries: Pandas, NumPy, Scikit-learn, NLTK, Matplotlib, Seaborn

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Tuning, Hyperparameter Optimization

Natural Language Processing: NLTK, Text Preprocessing, Tokenization, Sentiment Analysis, Text Classification

Computer Vision: OpenCV, Image Processing, Object Detection, Edge Detection, Image Transformation

Data Visualization: Power BI, Tableau, Streamlit, Jupyter Notebook, Google Colab, Excel

Frameworks and Tools: Flask, Streamlit, Git, GitHub, VS Code, Cisco Packet Tracer, Android Studio

Databases: MySQL, SQL

Mobile Development: Flutter, Dart

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern | Sprints | July 2025 - September 2025

- Studied applied AI concepts and practical machine learning implementations with focus on building end-to-end machine learning workflows
- Trained on data preprocessing, model training, validation, and performance tuning for robust predictive systems
- Implemented supervised machine learning algorithms including Logistic Regression and Random Forest to solve real-world problems
- Practiced model deployment concepts using API-based structures and Python automation scripts

Python for Data Analysis Intern | National Telecommunication Institute (NTI) | November 2024 - February 2025

- Performed in-depth exploratory data analysis (EDA) on diverse datasets to uncover critical trends and correlations
- Applied advanced statistical analysis and Python-based data processing techniques to extract meaningful insights
- Built compelling data visualizations using Matplotlib and Seaborn for comprehensive reporting and insight generation
- Prepared clean, high-quality datasets for machine learning pipelines through meticulous preprocessing and feature engineering
- Organized large volumes of raw data into structured, usable formats to facilitate informed decision-making
- Created structured reports summarizing analytical findings and providing actionable recommendations

Machine Learning for Data Analysis Training | National Telecommunication Institute (NTI) | August 2024 - September 2024

- Built supervised machine learning models for both classification and regression tasks addressing various data challenges
- Conducted comprehensive data preparation including feature scaling, encoding, and transformation for optimal model performance

- Actively participated in collaborative AI challenges and group-based machine learning projects fostering teamwork and problem-solving skills
- Gained practical exposure to real business problems and learned how AI solutions are effectively applied to solve them
- Strengthened proficiency in Python, core machine learning algorithms, and experimentation using Jupyter and Google Colab

AI and Machine Learning Intern | Orange Digital Center (AMIT Learning) | November 2024 - December 2024

- Applied artificial intelligence and machine learning concepts in practical projects gaining hands-on experience with real-world scenarios
- Worked extensively with supervised learning algorithms using Python implementing various models
- Implemented classification and regression models on diverse real-world datasets focusing on accuracy and efficiency

Data Analyst Intern | Creativa Hub Benha | September 2024 - October 2024

- Cleaned and transformed complex datasets to significantly improve data reliability and analytical quality
- Designed and developed interactive Power BI and Excel dashboards to support strategic business decision-making
- Built impactful visualizations and reports translating raw data into clear, actionable insights

PROJECTS

Sentiment Analysis App (Natural Language Processing and Machine Learning)

- Developed real-time Sentiment Analysis application using Python and Streamlit to classify text as Positive or Negative
- Implemented NLTK for text preprocessing including tokenization and stop-word removal
- Applied Scikit-learn for machine learning model training and prediction
- Designed user-friendly interface to provide businesses with instant insights into customer feedback

EarthlyEats Warehouse Dashboard (Freelance, National Telecommunication Institute Supervised)

- Developed comprehensive business dashboard for startup (EarthlyEats) under supervision of National Telecommunication Institute (NTI)
- Analyzed complex warehouse datasets to track inventory milestones and project delivery progress
- Delivered data-driven visualizations to optimize supply chain operations and decision-making

Red Object Detection System (Computer Vision)

- Built real-time object detection system specifically tuned for identifying red objects using OpenCV and Python
- Applied image processing techniques including color masking, contour detection, and morphological transformations

Cartoon Image Transformation (Computer Vision)

- Created image processing tool that converts standard photographs into cartoon-style illustrations
- Utilized OpenCV for edge detection using Laplacian and Canny filters and bilateral filtering

Airline Passenger Satisfaction Prediction (Machine Learning)

- Engineered full machine learning pipeline to predict passenger satisfaction with 90 percent plus accuracy
- Used Random Forest and Gradient Boosting algorithms for model training
- Applied hyperparameter tuning and cross-validation techniques to identify key satisfaction drivers

Heart Disease Prediction Model (Machine Learning)

- Developed diagnostic classification model using Support Vector Machine (SVM) and Logistic Regression
- Conducted feature correlation analysis to identify medically significant risk factors from clinical datasets

EDUCATION

Bachelor of Computer Science | Thebes Academy | Graduation: 2024

Grade Point Average: 3.16 out of 4.0 | Graduation Project: Excellent

CERTIFICATIONS AND TRAINING

- Machine Learning for Data Analysis - National Telecommunication Institute (NTI)
- Python for Data Analysis - National Telecommunication Institute (NTI)
- Artificial Intelligence - Orange Digital Center and AMIT Learning
- Data Analysis - Creativa Hub Benha
- Artificial Intelligence - Orange Digital Center Digital HUB (D-HUB)
- InnovEgypt Internship Program - ITIDA and TIEC
- Artificial Intelligence - Thebes Academy
- Networking - Thebes Academy
- Blockchain Basics - EYouth
- Flutter Development - Thebes Academy

KEY ACHIEVEMENTS

- Kaggle Expert: Achieved Kaggle Expert level with multiple published notebooks covering classification, regression, time series analysis, and advanced machine learning techniques
- Graduation Project: Completed excellent-rated graduation project demonstrating advanced technical and project management skills
- National Telecommunication Institute Training: Completed comprehensive training programs in machine learning and data analysis at leading national institution
- Internship Experience: Completed nine internships across data science, machine learning, data analysis, networking, and mobile development domains

- Project Portfolio: Developed and deployed six major projects in machine learning, natural language processing, and computer vision

LANGUAGES

Arabic: Native | English: Fluent